

Frank Tinsley

Interactive Systems & Real-Time Graphics Engineer · Unity · C# · GLSL / Metal · GPU

Portland, OR — onsite or hybrid locally; remote anywhere

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SUMMARY

Real-time graphics engineer with 15+ years building interactive systems, including a decade of sensor- and touch-driven installations that ran unattended in retail and public venues. Most recently contract Technical Artist at Meta (via Insight Global) on Meta Avatars, focused on real-time interactivity and rendering.

Solo-built an SDF ray-marching 3D modeling engine in Swift + Metal — ~50,000 lines across ~650 commits — running at interactive frame rates on iOS, iPadOS, and macOS. Prototypes broad and fast, then hardens until the frame time holds on the machine the work actually ships on.

SKILLS

- **Interactive installations:** experiences built for physical space and unattended public operation — custom embedded sensor hardware (Arduino) driving real-time 3D, markerless body tracking (Kinect), multi-user simultaneous interaction, multi-touch, large-format displays
- **Real-time rendering & shaders:** shader authoring and optimization (Metal MSL, HLSL/ShaderLab, GLSL / WebGL2), GPU compute, real-time rendering pipelines, graphics pipeline optimization, SDF / implicit surfaces, ray marching, procedural geometry, marching cubes
- **Engines & languages:** Unity, C#, Swift, SwiftUI, Metal, TypeScript/JavaScript, Go, Java
- **VR/AR/XR:** avatars, ARKit, Meta Spark AR, Lens Studio, mixed reality; cross-platform iOS/iPadOS/macOS and Android
- **Tooling & verification:** file-format and pipeline tooling (created the ARX AR interchange format); automated frame-diff verification against reference implementations; AI-accelerated development workflows

EXPERIENCE

Independent Software Developer — Self-Employed · Portland, OR · May 2023 – Present

Code and live demo available on request.

- Designed and built, solo, a real-time signed-distance-field (SDF) ray-marching 3D modeling engine and app in Swift + Metal (Feb–May 2026), running across iOS, iPadOS, and macOS.
- Authored the full shader stack — Metal compute, fragment, and vertex pipelines — for ray-marched implicit surfaces, smooth booleans and organic blending, and procedural geometry, optimized for interactive frame rates on mobile and desktop GPUs.
- Implemented marching-cubes mesh export (STL/OBJ), bridging the implicit-surface renderer into standard 3D content pipelines.
- Built a personal site that is itself a live WebGL2 SDF renderer — the page background is one signed-distance field solved per pixel at full device resolution, with no textures and no libraries.
- Designed and implemented an FPGA re-implementation of the Konami System 573 arcade platform for MiSTer (github.com/fabricore-eng/System573_MiSTer) — ~22,000 lines of custom RTL in about one month, in a domain with no prior background, built through an AI-accelerated workflow and proven frame-accurate against the MAME reference emulator via automated frame-diff tooling on real FPGA hardware.

Technical Artist — Meta (contract via Insight Global) · Portland, OR (onsite) · Mar 2025 – Feb 2026

- Designed and built a visual in-editor tool for authoring mock events and controlling them at runtime, enabling full behavioral testing of the live avatar systems and continual evolution of Meta's internal

live-avatars API.

- Worked on Meta Avatars — Meta's avatar system for VR and mixed-reality platforms — focused on real-time interactivity and avatar rendering.

AR Subject Matter Expert — Meta (contract via IntraEdge) • Mar 2022 – May 2023

- Served as Meta's subject matter expert on Spark AR Studio and AR development more broadly; created hands-on activities and tutorials for an educational curriculum training new AR creators.

Mixed Reality Developer — Microsoft • Remote • Sep 2021 – Mar 2022

Runtime Engineer — Camera IQ • Santa Monica, CA • Jan 2018 – Nov 2020

- Built key components of the Camera IQ platform's runtime and backend, starting in C# for the Unity runtime, then JavaScript for additional AR runtimes.
- Implemented backend services first in Java (with Swift for UI automation), then in Go — while continuing to develop new runtime features in JavaScript — including a Go wrapper covering nearly the entire Spark AR Studio file format.
- Created ARX, an API-neutral AR experience file format for interchange between AR runtimes including Snapchat's Lens Studio and Facebook's Spark AR Studio.

Principal / Contract Unity Developer — CodeStep LLC (independent consultancy) • Tigard, OR • Nov 2009 – Jan 2018

Ran the consultancy; delivered interactive experiences in Unity for mobile, XR, and branded retail installations, owning the entire creative and technical development process end to end.

- **Instrumented product → real-time digital twin** (major athletic shoe brand, in-store): built a real shoe with an Arduino inside it that streamed live motion data to a custom Unity application. Shoppers picked up the physical shoe and rotated it in their hands; that exact orientation drove a real-time render of the same shoe on screen, surfacing product features as different parts turned toward the camera. Exhibited at trade shows worldwide and in retail.
- **Multiplayer body-tracked game** (Coca-Cola, Disney World): a Microsoft Kinect installation letting multiple simultaneous players perform swimming motions to race each other as polar bears, in a Coca-Cola-branded Unity game on a large screen at the Coca-Cola store at Disney World.
- **Multi-touch frost wall** (Coca-Cola): a large touchscreen that appeared to frost over; guests touched to wipe the frost away, revealing a window into an aquarium with life-size swimming polar bears.
- Built original photo-booth software installed in retail locations.
- Developed the game "Fascination," purchased by GSN, then worked with GSN's team to improve it; built Unity game systems for the cross-platform mobile wellness game "Habit Monster."